

Day 9 of training

OPERATORS IN JAVASCRIPT

there are six types of operators

1). arithmetic operators

2). logical operators

3). comparison operators

4). assignment operators

5). string operators

6). ternary operators

1. Arithmetic Operators (Math stuff)

Operator	Meaning	Example
-----	-----	-----
`+`	Addition	`5 + 3 = 8`
`-`	Subtraction	`5 - 3 = 2`
`*`	Multiplication	`5 * 3 = 15`
`/`	Division	`6 / 2 = 3`
`%`	Modulus (remainder)	`5 % 2 = 1`
`**`	Exponentiation	`2 ** 3 = 8`
`++`	Increment	`x++` means `x = x + 1`
`--`	Decrement	`x--` means `x = x - 1`

2. Assignment Operators

Operator	Meaning	Example
-----	-----	-----
`=`	Assign value	`x = 5`
`+=`	Add and assign	`x += 2` → `x = x + 2`
`-=`	Subtract and assign	`x -= 2`

`*=`	Multiply and assign	`x *= 2`	
`/=`	Divide and assign	`x /= 2`	
`%=`	Modulus and assign	`x %= 2`	

3. Comparison Operators (Used in conditions)

Operator	Meaning	Example	
-----	-----	-----	
`==`	Equal to (value only)	`5 == "5" → `true`	
`===`	Equal to (value and type)	`5 === "5" → `false`	
`!=`	Not equal (value only)	`5 != "5" → `false`	
`!==`	Not equal (value and type)	`5 !== "5" → `true`	
`>`	Greater than	`5 > 3`	
`<`	Less than	`5 < 3`	
`>=`	Greater than or equal to	`5 >= 5`	
`<=`	Less than or equal to	`3 <= 5`	

4. Logical Operators

Operator	Meaning	Example				
-----	-----	-----	--	-----	-	-----
`&&`	AND	`true && false` → `false`				
`  `	OR	`true    false` → `true`				
`!`	NOT	`!true` → `false`				

5. String Operator

+ → Also used to concatenate (join) strings

"Hello" + " Opinder" // "Hello Opinder"

6.Ternary Operator (Shortcut for if-else)

```
condition ? trueResult : falseResult;
```

```
let result = (age >= 18) ? "Adult" : "Minor";
```

Bitwise AND Truth Table

A	B	A & B
---	---	-------

-	-	-----
---	---	-------

0	0	0
---	---	---

0	1	0
---	---	---

1	0	0
---	---	---

1	1	1
---	---	---

Bitwise OR Truth Table

A	B	A B
---	---	-------

-	-	-----
---	---	-------

0	0	0
---	---	---

0	1	1
---	---	---

1	0	1
---	---	---

1	1	1
---	---	---

Bitwise XOR Truth Table

A	B	A ^ B
---	---	-------

-	-	-----
---	---	-------

0	0	0
---	---	---

0	1	1
---	---	---

1	0	1
---	---	---

1	1	0
---	---	---

What is Bitwise NOT (~)?

It flips each bit of the number:

0 becomes 1

1 becomes 0

| A | ~A |

| - | --- |

| 0 | 1 |

| 1 | 0 |

What is an if statement?

It's a decision maker.

It checks if a condition is true — and if it is, it runs the code inside the { }.

Syntax:

```
if (condition) {  
    // code runs if condition is true  
}
```

What is if...else?

if checks a condition

else runs if the condition is false

Syntax:

```
if (condition) {  
    // runs if condition is true  
} else {  
    // runs if condition is false  
}
```

In JavaScript, `if...else if...else` is used to check multiple conditions.

Here's the basic syntax:

```
if (condition1) {  
    // code runs if condition1 is true  
} else if (condition2) {  
    // code runs if condition2 is true  
} else {  
    // code runs if none of the above are true  
}
```